# MoveOn Phone Cover Pipeline & Inventory Synchronization Standard

**Specification Version:** 3.0 | **Project:** inventory/coverstore | **Status:** Production

## 1. System Architecture & Scope

Automates single-product asset acquisition from **moveon.global** with zero out-of-scope crawling. Features automatic material identification, interactive phone model selection, standardized SKU generation, **in-place image translation** replacing original non-English pictures, and automated inventory sync.

## 2. Custom SKU Generation Standard

All variation cover images and database records are named using the standardized format:
**[model abbreviation]-[material abbreviation]-[color abbreviation]-[print abbreviation]**
*Example:* RN8-SIL-WT-S.jpg (Realme Neo8, Liquid Silicone, Antique White, Solid Color)

| Component | Code Format | Representative Examples |
|---|---|---|
| Phone Model | 2-6 Alphanumeric | `RN8 (Realme Neo8), IP15PM (iPhone 15 Pro Max), S24U (Galaxy S24 Ultra)` |
| Material | 3 Letters | `SIL (Liquid Silicone), GLS (Glass), TPU (TPU), LTH (Leather), PC (Hard PC)` |
| Color | 2 Letters | `WT (White), BK (Black), PR (Purple), GN (Green), BL (Blue), CL (Clear)` |
| Print / Finish | 1 Letter | `S (Solid/Cube), P (Printed), G (Gradient), E (Electroplated/Mirror), C (Clear)` |

## 3. Storage Hierarchy (Strict 2-Folder Structure)

Assets are organized into exactly two directories per model and material. Non-English text is translated in-place:

| Directory / File | Description & Specification |
|---|---|
| `\[model]\[material]\gallery\` | gallery_01.jpg ... Product showcase images. Non-English text translated in-place. |
| `\[model]\[material]\sku\` | [SKU].jpg (e.g. RN8-SIL-WT-S.jpg) ... Variant covers translated in-place. |
| `\inventory.csv & .xlsx` | 13-column database synchronized with Google Sheet (gid=641365888). |
| `\scripts\` | Modules: downloader.py, inventory_manager.py, sku_generator.py, etc. |

## 4. Inventory Dataset Standard (13-Column Schema)

Synchronizes live Google Sheet rows, enriches each variant via MoveOn wholesale API, and records local assets:

| Col # | Field Name | Data Description & Source | Realme Neo8 Example |
|---|---|---|---|
| 1 | Product Name | MoveOn product title | `Suitable for Realme Neo8 Phone Case...` |
| 2 | Product Code | Standardized custom SKU | `RN8-SIL-WT-S` |
| 3 | Material | Material identified | `Liquid Silicone` |
| 4 | Color | Variant color label | `Antique White` |
| 5 | Phone Model | Selected phone model | `Realme Neo8` |
| 6 | Warranty | Warranty status | `No` |
| 7-8 | source link / link | MoveOn product URL | `https://moveon.global/BD_en/products/...` |
| 9 | Price (CNY) | Dropship / wholesale price | `5.0` |
| 10 | Stock | Available variant inventory | `999219` |
| 11 | Weight (kg) | Single SKU shipping weight | `0.03` |
| 12 | Image URL | Remote CDN variant image | `https://img.alibabaim.com/...` |
| 13 | Local Image Path | Local relative asset path | `Realme Neo8/Liquid Silicone/sku/RN8-SIL-WT-S.jpg` |